Derivation of the Modified Fresnel Coefficients

Based on Bedeaux and Vlieger's Formalism

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1 Introduction

This section describes the derivation of modified Fresnel coefficients, following the approach of Bedeaux and Vlieger [1]. We begin with Ampere's circuital law:

$$\nabla \times \mathbf{H} = \frac{1}{c}\mathbf{I} + \frac{1}{c}\frac{\partial \mathbf{D}}{\partial t}, \tag{1}$$

where $\mathbf{H}$ and $\mathbf{D}$ are the magnetic and electric displacement fields, $\mathbf{I}$ is the current density, and c is the speed of light.

2 Boundary Conditions and Field Decomposition

We consider a boundary in the xy-plane and decompose the fields:

$$\mathbf{H} = \mathbf{H}^{+}\theta(z) + \mathbf{H}^{-}\theta(-z) + \mathbf{H}^{s}\delta(z), \tag{2}$$

$$\mathbf{D} = \mathbf{D}^{+}\theta(z) + \mathbf{D}^{-}\theta(-z) + \mathbf{D}^{s}\delta(z), \tag{3}$$

where $\theta(z)$ and $\delta(z)$ are the Heaviside and Dirac delta functions, respectively. The fields $\mathbf{H}^s$ and $\mathbf{D}^s$ satisfy $D_{xs} = D_{ys} = H_{xs} = H_{ys} = 0$.

Applying Ampere's law at $z = 0$, we find:

$$\nabla \times \mathbf{H}_s^z + \hat{z} \times (\mathbf{H}_{||}^{+} - \mathbf{H}_{||}^{-})\delta(z) = \frac{1}{c}\mathbf{I}_s + \frac{1}{c}\frac{\partial \mathbf{D}_s}{\partial t}. \tag{4}$$

Multiplying by $(\hat{z}\times)$ and rearranging, we obtain:

$$\mathbf{H}_{||}^{+} - \mathbf{H}_{||}^{-} = -\hat{z} \times \frac{1}{c}\mathbf{I}_{||}^{s} - \hat{z} \times \frac{1}{c}\frac{\partial \mathbf{P}_{||}^{s}}{\partial t} - \nabla_{||} M_z^s, \tag{5}$$

where $\mathbf{P}_s$ and $\mathbf{M}_s$ represent the electric and magnetic polarization densities.

Defining a generalized displacement field $\mathbf{D}' = \mathbf{D} + i/\omega \mathbf{I}$ and assuming no magnetic polarization at the boundary, we obtain:

$$\mathbf{H}_{||}^{+} - \mathbf{H}_{||}^{-} = \frac{i\omega}{c}\hat{z} \times \mathbf{P}_{||}^{s}. \tag{6}$$

3 Modified Reflection and Transmission Coefficients

For s-polarized light, we assume polarization along y, leading to:

$$\mathbf{P}_{||} = -\rho\alpha(\omega)E_{exc,x}, \tag{7}$$

where $\alpha(\omega)$ is the polarizability of a single nanoparticle and ρ is the surface density of nanoparticles. The exciting field is:

$$E_{exc,y} = E_t = (E_i + E_r). \tag{8}$$

By applying boundary conditions for the magnetic field:

$$H_x^- = n_i(-E_i + E_r)\cos\theta_i, \tag{9}$$

$$H_x^+ = -n_t E_t \cos\theta_t = -n_t(E_i + E_r)\cos\theta_t. \tag{10}$$

Inserting these into the boundary condition, we derive the modified reflection coefficient:

$$r_s = \frac{n_i\cos\theta_i - n_t\cos\theta_t + i\frac{\omega}{c}\rho\alpha(\omega)}{n_i\cos\theta_i + n_t\cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)}. \tag{11}$$

For transmission, using $t_s = r_s + 1$:

$$t_s = \frac{2n_i\cos\theta_i}{n_i\cos\theta_i + n_t\cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)}. \tag{12}$$

The same procedure can be applied for p-polarized light, yielding:

$$r_p = \frac{n_t\cos\theta_i - n_i\cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)\cos\theta_i\cos\theta_t}{n_t\cos\theta_i + n_i\cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)\cos\theta_i\cos\theta_t}. \tag{13}$$

And for transmission:

$$t_p = \frac{2n_i\cos\theta_i}{n_t\cos\theta_i + n_i\cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)\cos\theta_i\cos\theta_t}. \tag{14}$$

4 Conclusion

We have derived the modified Fresnel coefficients by incorporating nanoparticle polarizability and surface density into boundary conditions. This formulation extends classical Fresnel equations and provides a framework for understanding light interaction with structured interfaces.

References

[1] Bedeaux, D., and Vlieger, J. *Optical Properties of Surfaces.* Imperial College Press, 2001.

5 Supporting Information

5.1 Derivation of the Modified Fresnel Coefficients

This section aims to describe the origin of Eq. (3,4) in the main paper. We follow Bedeaux and Vlieger [1], to which we refer the reader for a more extensive derivation.

To derive the modified Fresnel coefficients, it is convenient to start from Ampere's circuital law, which may be written as:

$$\nabla \times \mathbf{H} = \frac{1}{c}\mathbf{I} + \frac{1}{c}\frac{\partial \mathbf{D}}{\partial t}, \tag{15}$$

where $\mathbf{H}$ and $\mathbf{D}$ are the magnetic and electric displacement fields, $\mathbf{I}$ is the current density, and c is the speed of light.

Considering a boundary in the xy-plane, we may divide the $\mathbf{H}$ and $\mathbf{D}$ fields into three components:

$$\mathbf{H} = \mathbf{H}^{+}\theta(z) + \mathbf{H}^{-}\theta(-z) + \mathbf{H}^{s}\delta(z), \tag{16}$$

$$\mathbf{D} = \mathbf{D}^{+}\theta(z) + \mathbf{D}^{-}\theta(-z) + \mathbf{D}^{s}\delta(z), \tag{17}$$

where $+$, $-$, and s denote the fields above, below, and within the boundary. $\theta(z)$ and $\delta(z)$ are the Heaviside and Dirac delta functions. It is convenient to choose $\mathbf{D}^s$ and $\mathbf{H}^s$ such that $D_{xs} = D_{ys} = H_{xs} = H_{ys} = 0$, without any loss in generality.

Using Ampere's circuital law at $z = 0$, and noting that $\partial\theta(\pm z)/\partial z = \pm\delta(z)$, it can be shown that:

$$\nabla \times \mathbf{H}_s^z + \hat{z} \times (\mathbf{H}_{||}^{+} - \mathbf{H}_{||}^{-})\delta(z) = \frac{1}{c}\mathbf{I}_s + \frac{1}{c}\frac{\partial \mathbf{D}_s}{\partial t}. \tag{18}$$

Multiplying by $(\hat{z}\times)$ and making some simple rearrangements, one finds at $z = 0$:

$$\mathbf{H}_{||}^{+} - \mathbf{H}_{||}^{-} = -\hat{z} \times \frac{1}{c}\mathbf{I}_{||}^{s} - \hat{z} \times \frac{1}{c}\frac{\partial \mathbf{P}_{||}^{s}}{\partial t} - \nabla_{||} M_z^s. \tag{19}$$

Here, $\mathbf{P}_{||}^{s} = \mathbf{D}_{||}^{s}$ and $M_z^s = -H_s^z$ are the electric and magnetic polarizations, and $\nabla_{||} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right)$.

By defining a generalized electric displacement field $\mathbf{D}' = \mathbf{D} + i/\omega\mathbf{I}$, the electric current is absorbed in $\mathbf{D}'$. Assuming no magnetic polarization in the boundary, this leads us to the boundary condition:

$$\mathbf{H}_{||}^{+} - \mathbf{H}_{||}^{-} = \frac{i\omega}{c}\hat{z} \times \mathbf{P}_{||}^{s}. \tag{20}$$

For s-polarization, where the polarization is along y, we assume:

$$\mathbf{P}_{||} = -\rho\alpha(\omega)E_{exc,x}, \tag{21}$$

where $\alpha(\omega)$ is the polarizability of a single nanoparticle and ρ is the surface density of nanoparticles. The exciting field is:

$$E_{exc,y} = E_t = (E_i + E_r). \tag{22}$$

Applying boundary conditions for the magnetic field:

$$H_x^- = n_i(-E_i + E_r)\cos\theta_i, \tag{23}$$

$$H_x^+ = -n_t E_t \cos\theta_t = -n_t(E_i + E_r)\cos\theta_t. \tag{24}$$

By combining these relationships, we find:

$$n_i \cos\theta_i(E_i - E_r) - n_t \cos\theta_t(E_i + E_r) = -\frac{i\omega}{c}\rho\alpha(\omega)(E_i + E_r). \tag{25}$$

Rearranging, we obtain the modified reflection coefficient:

$$r_s = \frac{n_i \cos\theta_i - n_t \cos\theta_t + i\frac{\omega}{c}\rho\alpha(\omega)}{n_i \cos\theta_i + n_t \cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)}. \tag{26}$$

For transmission:

$$t_s = \frac{2n_i \cos\theta_i}{n_i \cos\theta_i + n_t \cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)}. \tag{27}$$

The same procedure applies for p-polarized light, yielding:

$$r_p = \frac{n_t \cos\theta_i - n_i \cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)\cos\theta_i\cos\theta_t}{n_t \cos\theta_i + n_i \cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)\cos\theta_i\cos\theta_t}. \tag{28}$$

And for transmission:

$$t_p = \frac{2n_i \cos\theta_i}{n_t \cos\theta_i + n_i \cos\theta_t - i\frac{\omega}{c}\rho\alpha(\omega)\cos\theta_i\cos\theta_t}. \tag{29}$$